ROYAL FITNESS CLUB

RFC PREMIUM GUIDE #08

TRT HORMONE OPTIMISATION GUIDE

Complete Protocol for Testosterone Replacement Therapy

5 Chapters · Medical Reference · Monitoring Protocols · Safety First

TABLE OF CONTENTS

DISCLAIMER

This guide is for educational and harm-reduction purposes only. It does not constitute medical advice. Consult a qualified healthcare professional before starting any supplementation or pharmaceutical protocol. Individual results vary.

CHAPTER 1 — UNDERSTANDING LOW TESTOSTERONE

"Low testosterone is not an inevitable part of ageing — it is a treatable medical condition with profound quality-of-life implications when left unaddressed."

Testosterone Replacement Therapy (TRT) is a medically supervised protocol for men with clinically low testosterone levels (hypogonadism). Low T is defined as serum testosterone below 300 ng/dL (10.4 nmol/L) by most guidelines, accompanied by symptoms. In India, lifestyle factors, stress, poor sleep, metabolic dysfunction, and environmental exposures have led to a significant increase in younger men presenting with suboptimal testosterone levels.

Symptom Category	Common Symptoms	Frequency
Physical	Reduced muscle mass, increased body fat, fatigue, decreased strength	Very common
Sexual	Low libido, erectile dysfunction, reduced morning erections	Common
Psychological	Depression, irritability, poor concentration, lack of drive	Common
Metabolic	Insulin resistance, elevated triglycerides, central adiposity	Moderate
Sleep	Poor sleep quality, sleep apnoea correlation	Moderate

Diagnostic Criteria

Lab Test	Optimal Range (TRT Candidate)	Action if Below Range
Total Testosterone	600–1000 ng/dL optimal; < 300 = clinical low	TRT evaluation with endocrinologist
Free Testosterone	15–25 pg/mL optimal	SHBG investigation
SHBG	20–40 nmol/L	If high: reduces free T — may need TRT even with normal total T
LH / FSH	If low: secondary hypogonadism	HCG may restore natural production
Estradiol	20–30 pg/mL	Aromatase inhibitor may be needed if elevated

CHAPTER 2 — TRT PROTOCOLS

Method	Dose	Frequency	Pros	Cons
Testosterone Cypionate (IM)	100–200mg/wk	Weekly or biweekly	Stable levels, affordable	Injection required, fluctuation
Testosterone Enanthate (IM)	100–200mg/wk	Weekly or biweekly	Same as Cypionate	Slightly more E2 fluctuation
Testosterone Propionate (IM)	50mg	Every other day	Stable levels, less conversion	Frequent injections
Testosterone Gel (topical)	50–100mg/day	Daily	No injections	Transference risk, absorption varies
Testosterone Pellets	150–450mg	Every 3–6 months	Convenience, stable	Insertion procedure required

■ ***Most TRT specialists recommend starting at 100mg/week and adjusting based on blood work at 8 weeks. Avoid starting high — it is easier to increase than to lower a dose that has caused side effects.***

CHAPTER 3 — MONITORING & OPTIMISING

TRT Monitoring Schedule

Timepoint	Tests Required	What to Assess
Baseline (pre-TRT)	Full hormonal panel + CBC + CMP + PSA + lipids	Establish starting point for all markers
Week 6–8	Total T, Free T, E2, hematocrit, PSA	Dose adequacy and estrogen management
Week 16	Full panel repeat	Comprehensive optimisation review
Every 6 months (stable)	Full panel + PSA + bone density (annual)	Long-term safety monitoring
Annually	Full panel + PSA + cardiovascular risk	Comprehensive annual review

Optimisation Targets on TRT

Marker	Target Range on TRT	Adjustment
Total Testosterone	700–1000 ng/dL	Increase dose if < 600; reduce if > 1200
Estradiol (E2)	20–35 pg/mL	AI if consistently > 40 pg/mL
Hematocrit	< 52%	Donate blood or reduce dose if > 52%
PSA	< 4 ng/mL (age-adjusted)	If rise > 1.4 in 1 year, urology referral
Blood Pressure	< 130/80	Lifestyle, dose adjustment, or medication

CHAPTER 4 — MANAGING SIDE EFFECTS

Side Effect	Cause	Management	When to Act Urgently
Acne	Elevated androgens	Salicylic acid, adapalene, doxycycline	Cystic/scarring acne
Hair thinning	DHT conversion	Nizoral shampoo, finasteride (Rx)	If distressing to patient
Elevated hematocrit	RBC stimulation	Donate blood, hydrate, reduce dose	If > 54%
Gynecomastia	Elevated E2	Anastrozole 0.25–0.5mg twice weekly	Hard gland tissue developing
Testicular atrophy	LH/FSH suppression	HCG 250IU twice per week	If fertility desired
Mood changes	Hormonal fluctuation	Split injections to stabilise levels	Severe depression or aggression

■ ***HCG co-administration preserves testicular function and natural testosterone production during TRT. Recommended dose: 250–500 IU subcutaneously twice per week.***

CHAPTER 5 — TRT & FITNESS

"TRT brings testosterone to a healthy physiological range — it is not performance enhancement, it is physiological restoration. The fitness benefits are significant nonetheless."

Fitness Parameter	Expected Change on TRT	Timeline	Optimisation Strategy
Lean muscle mass	+2–5kg over 12 months	3–6 months onset	Resistance training 4 days/week
Body fat	-2–4% total body fat	3–6 months	Combined deficit + resistance training
Strength	+10–20% on compound lifts	2–4 months	Progressive overload protocol
Energy / libido	Significant improvement	4–6 weeks	Sleep, stress management
Bone density	Gradual improvement	12–24 months	Weight-bearing exercise

- Training on TRT: treat as enhanced recovery — 4–5 resistance sessions per week is optimal
- Protein target: 2.0–2.5g per kg bodyweight (enhanced utilisation on TRT)
- Cardiovascular exercise critical: TRT increases RBC and can elevate BP — 3× cardio/week mandatory
- Sleep quality directly impacts TRT efficacy — target 7–8 hours; address sleep apnoea